

C891

Ecology and Environmental Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A deer herd on a reserve managed by a nonprofit has a carrying capacity of 800; biologists count it using drones that cost \$12,000. At what herd size is the number of deer added per year greatest, assuming logistic growth?

- A. Very small sizes, where the growth percentage is highest
- B. The same at every size below 800
- C. 400
- D. 800

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A population following logistic growth adds individuals fastest when it has 1,500 members. What is its carrying capacity?

- A. 1,500
- B. It cannot be determined without birth rates.
- C. 3,000
- D. 750

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A frog population in a pond, surveyed by volunteers in bright green shirts, drops sharply after a late frost that kills the same fraction of frogs whether the pond is crowded or not. What kind of limiting factor is the frost?

- A. Density-independent
- B. A factor that raises carrying capacity because fewer frogs compete
- C. A biotic factor
- D. Density-dependent

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Barnacles attach to a whale and ride to food-rich waters; the whale is neither helped nor harmed. The whale was tagged by a research vessel from a coastal university. What relationship is this?

- A. Commensalism
- B. Parasitism
- C. Competition
- D. Mutualism

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which relationship has one species benefiting while the other is harmed but usually not killed right away?

- A. Mutualism
- B. Predation
- C. Commensalism
- D. Parasitism

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Two paramecium species, ordered from the same supply catalog, thrive in separate tanks. Grown together with one shared food source, one species dies out. Which principle explains this?

- A. Predation
- B. Mutualism
- C. Competitive exclusion
- D. Species with the same food always coexist if food is plentiful.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Five similar warbler species eat insects in the same spruce trees yet coexist. What best explains this?

- A.** They share identical niches with no effect on each other.
- B.** Niche partitioning: each feeds in a different zone of the tree.
- C.** Competition never occurs among birds.
- D.** They are mutualists that feed each other.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a tide-pool community photographed for a nature magazine, removing a predatory sea star that makes up under 5% of the biomass lets mussels take over, and the number of species drops from 15 to 8. How is the sea star best described?

- A.** A keystone species
- B.** Unimportant, since it made up so little biomass
- C.** A dominant species
- D.** An invasive species

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

After a wildfire burns part of a national park popular with campers, the soil remains and grasses return within a year. What process is this?

- A.** A climax community
- B.** Primary succession
- C.** No succession, because fire destroys all life
- D.** Secondary succession

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A team from a university that recently won a conservation award tags 60 turtles and releases them. A month later they catch 50 turtles, and 10 have tags. What is the estimated population?

- A.** 3,000
- B.** 300
- C.** 12
- D.** 110

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A population is estimated at 400. Researchers had marked 80 animals and later caught 50. How many marked animals were in the second catch?

- A.** 5
- B.** 50, because every animal caught must be marked
- C.** 250
- D.** 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A weedy plant colonizing empty lots in a city with a new light-rail line makes thousands of tiny seeds, matures in weeks, and gives no care to offspring. Which life-history strategy is this?

- A.** K-selected
- B.** r-selected
- C.** Both equally, as all species balance the two
- D.** Keystone

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A deer herd on a reserve ~~managed by a nonprofit~~ has a carrying capacity of 800; ~~biologists count it using drones that cost \$12,000.~~ At what herd size is the number of deer added per year greatest, assuming logistic growth?

Target word: size at fastest growth **Decides:** Logistic growth peaks at $K/2$.

A	Adjacent category	Per-capita rate is highest there, but total additions are small.
B	Dropped element	Drops the slowing that defines logistic growth.
C	Correct	$800 \div 2$.
D	Confusion set	At K , growth is near zero.

2. Answer C Concept 1 • transfer

A population following logistic growth adds individuals fastest when it has 1,500 members. What is its carrying capacity?

Target word: solve for K **Decides:** Fastest growth at $K/2$.

A	Wrong stem number	That is $K/2$, not K .
B	Absolute language	The $K/2$ rule is enough.
C	Correct	$1,500 = K/2$.
D	Inverted operation	Halves instead of doubling.

3. Answer A Concept 2

A frog population in a pond, ~~surveyed by volunteers in bright green shirts,~~ drops sharply after a late frost that kills the same fraction of frogs whether the pond is crowded or not. What kind of limiting factor is the frost?

Target word: classify the factor **Decides:** Same fraction killed at any density.

A	Correct	Its effect does not depend on crowding.
B	Inverted operation	Lower numbers do not change the environment's capacity.
C	Adjacent category	Frost is abiotic.
D	Confusion set	That would hit crowded ponds harder.

4. Answer A Concept 3

Barnacles attach to a whale and ride to food-rich waters; the whale is neither helped nor harmed. ~~The whale was tagged by a research vessel from a coastal university.~~ What relationship is this?

Target word: symbiosis type **Decides:** + for barnacles, 0 for the whale.

A	Correct	One benefits; the other is unaffected.
B	Confusion set	Requires the host to be harmed.
C	Adjacent category	Both would be harmed by sharing a limited resource.
D	Confusion set	Requires both to benefit.

5. Answer D Concept 3 • transfer

Which relationship has one species benefiting while the other is harmed but usually not killed right away?

Target word: name from the effect pattern **Decides:** +/– with a living host.

A	Inverted operation	Both benefit; nobody is harmed.
B	Adjacent category	Also +/–, but the prey is killed and eaten.
C	Confusion set	The partner is unaffected.
D	Correct	The parasite feeds on a living host over time.

6. Answer C Concept 4

Two paramecium species, ~~ordered from the same supply catalog~~, thrive in separate tanks. Grown together with one shared food source, one species dies out. Which principle explains this?

Target word: name the principle **Decides:** Same niche, one limited resource.

A	Adjacent category	Neither species eats the other.
B	Inverted operation	Both would benefit, not one disappear.
C	Correct	The better competitor for the shared resource wins.
D	Absolute language	Plentiful food still runs short as populations grow.

7. Answer B Concept 4 • transfer

Five similar warbler species eat insects in the same spruce trees yet coexist. What best explains this?

Target word: why coexistence works **Decides:** They forage in different parts of the tree.

A	Inverted operation	Identical niches lead to exclusion.
B	Correct	Dividing the resource reduces direct competition.
C	Absolute language	Birds compete for food and nest sites.
D	Adjacent category	No mutual benefit is described.

8. Answer A Concept 5

In a tide-pool community ~~photographed for a nature magazine~~, removing a predatory sea star that makes up under 5% of the biomass lets mussels take over, and the number of species drops from 15 to 8. How is the sea star best described?

Target word: role classification **Decides:** Small biomass, large community effect.

A	Correct	Its effect far exceeds its abundance.
B	Inverted operation	Its removal cut species richness nearly in half.
C	Confusion set	Dominant species have the most biomass.
D	Adjacent category	Nothing says it is non-native.

9. Answer D Concept 6

After a wildfire burns part of a national park ~~popular with campers~~, the soil remains and grasses return within a year. What process is this?

Target word: succession type **Decides:** Soil remains.

A	Adjacent category	That is the late, stable stage, not early regrowth.
B	Confusion set	Starts on bare rock with no soil.
C	Absolute language	Seeds, roots, and soil organisms survive.
D	Correct	Regrowth on existing soil after disturbance.

10. Answer B Concept 7

A team ~~from a university that recently won a conservation award~~ tags 60 turtles and releases them. A month later they catch 50 turtles, and 10 have tags. What is the estimated population?

Target word: population estimate **Decides:** $N = 60 \times 50 \div 10$.

A	Dropped element	Forgot to divide by the 10 recaptures.
B	Correct	$3,000 \div 10$.
C	Inverted operation	$60 \times 10 \div 50$ flips the ratio.
D	Confusion set	Adds the two catches instead of using a proportion.

11. Answer D Concept 7 • transfer

A population is estimated at 400. Researchers had marked 80 animals and later caught 50. How many marked animals were in the second catch?

Target word: solve for recaptures **Decides:** $R = 80 \times 50 \div 400$.

A	Dropped element	$400 \div 80$ drops the second-catch size.
B	Absolute language	Only a fraction of any catch is marked.
C	Inverted operation	$400 \times 50 \div 80$ solves the wrong proportion.
D	Correct	$4,000 \div 400$.

12. Answer B Concept 8

A weedy plant colonizing empty lots ~~in a city with a new light rail line~~ makes thousands of tiny seeds, matures in weeks, and gives no care to offspring. Which life-history strategy is this?

Target word: strategy **Decides:** Many offspring, fast maturity, no care.

A	Confusion set	K-selected species have few, well-cared-for offspring.
B	Correct	Built for rapid colonization of disturbed habitats.
C	Absolute language	Species fall along a spectrum; this one is clearly r.
D	Adjacent category	A community role, not a life-history strategy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Logistic growth and carrying capacity	In logistic growth, the number added per year peaks at $K/2$ and slows toward zero as the population nears K .	Thinking growth is fastest near carrying capacity.	A carrying capacity and “when is growth fastest?”
2	Density-dependent vs. density-independent factors	Density-dependent factors (disease, competition, predation) hit harder as density rises; density-independent factors (frost, fire, floods) act regardless of density.	Calling any factor that reduces a population density-dependent.	Whether the same fraction dies at any population size.
3	Symbiosis types	Mutualism $+/+$, commensalism $+/0$, parasitism $+/-$.	Mixing up commensalism and mutualism, or parasitism and predation.	How each partner is affected: helped, harmed, or unaffected.
4	Competitive exclusion and niche partitioning	Two species cannot share the exact same niche indefinitely; coexisting competitors divide resources.	Assuming similar species coexist without dividing resources.	One species vanishes in a shared tank, or similar species feed in different zones.
5	Keystone species	A keystone species has an effect on community structure far larger than its abundance; removing it sharply changes diversity.	Equating importance with biomass or abundance.	A rare predator’s removal causes a big drop in species richness.
6	Primary vs. secondary succession	Primary succession starts on bare rock with no soil; secondary succession follows a disturbance that leaves soil.	Calling any regrowth after a big disturbance primary.	Whether soil remains.
7	Mark-recapture estimate	$N = (\text{number marked} \times \text{number in second catch}) \div \text{number of marked recaptures}$.	Inverting the ratio or adding the samples.	Tagged, released, and recaptured counts.
8	r- vs. K-selected life histories	r-selected species mature fast and make many offspring with little care; K-selected species mature slowly and invest heavily in few offspring.	Linking many offspring with stable, crowded habitats.	Offspring number, maturation time, and parental care.

Course scope (public description): Ecology and Environmental Science is an introductory course for graduate students seeking initial licensure or endorsement and/or to earn their MA degree in secondary or middle grade science education. The course explores the relationships between organisms and their environment, including population ecology, communities, adaptations, distributions, interactions, and the environmental factors controlling these relationships. This course has no prerequisites.